

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing(01CT1513)	Aim :- Perform gray level operations images.	
Lab-7	14-08-2026	Enrolment No:92400133131

Ami :- **Perform gray level operations images.**

Github link :- <https://github.com/kaladiyafarhan9879/Digital-Signal-and-Image-Processing.git>

```

Code :- import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image in grayscale
src_image = cv2.imread(
    "1.JPJ.jpg",
    cv2.IMREAD_GRAYSCALE
)

# Check if image is loaded
if src_image is None:
    print("Image not found!")
    exit()

# 1. Image Negation
negative_image = 255 - src_image

# 2. Thresholding
_, thresholded_image = cv2.threshold(
    src_image,
    128,
    255,
    cv2.THRESH_BINARY
)

# 3. Gamma Correction
gamma = 2.0

# Normalize image to range [0, 1]
normalized_image = src_image / 255.0

```

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# Apply gamma correction
gamma_corrected_image = np.power(
    normalized_image,
    1 / gamma
)

# Convert back to range [0, 255]
gamma_corrected_image = np.uint8(
    gamma_corrected_image * 255
)

# Display all images
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(src_image, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 4, 2)
plt.imshow(negative_image, cmap="gray")
plt.title("Negative")
plt.axis("off")

plt.subplot(1, 4, 3)
plt.imshow(thresholded_image, cmap="gray")
plt.title("Threshold")
plt.axis("off")

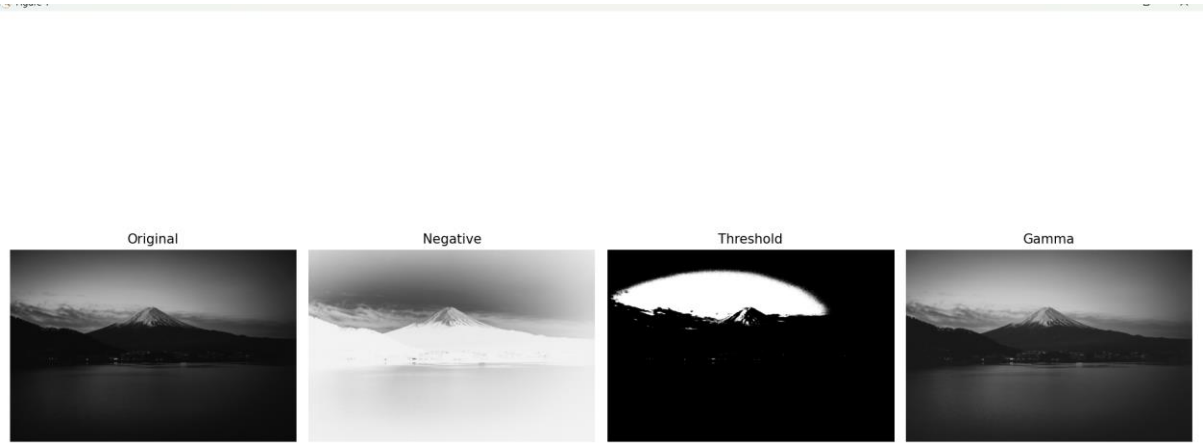
plt.subplot(1, 4, 4)
plt.imshow(gamma_corrected_image, cmap="gray")
plt.title("Gamma")
plt.axis("off")

plt.tight_layout()
plt.show()

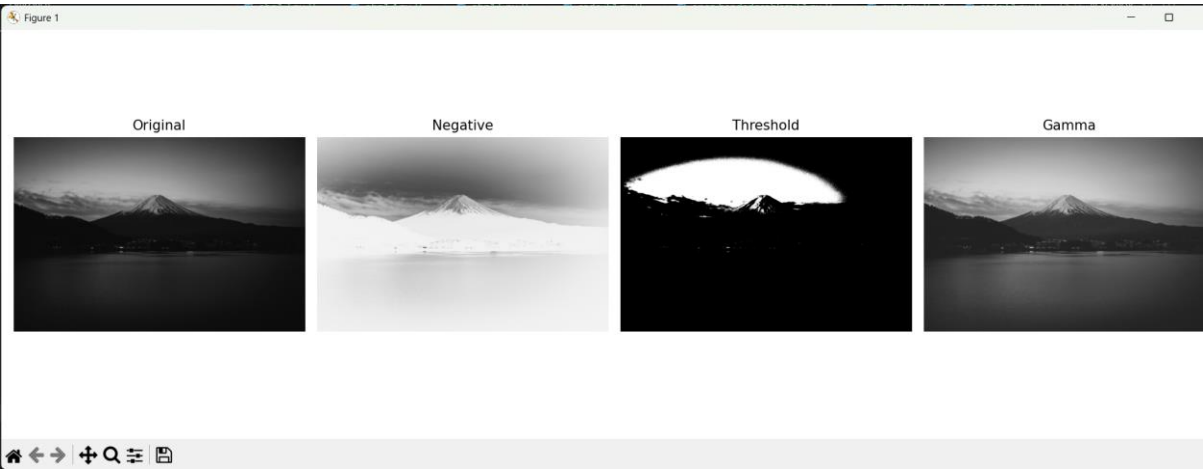
```

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OUTPUT :-



GAMMA =0.5



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Conclusion

1. The given Python program successfully performs basic image processing operations using OpenCV and NumPy.
2. The input image is first converted into grayscale for processing.
3. Image negation produces the complementary intensity of each pixel.
4. Thresholding converts the grayscale image into a binary image using a threshold value of 128.
5. Pixels below the threshold become black, while pixels above the threshold become white.
6. Gamma correction is applied to adjust the brightness and intensity of the image.
7. The image is normalized to the range 0 to 1 before applying the gamma transformation.
8. NumPy is used to efficiently perform the mathematical gamma correction operation.
9. The processed gamma image is converted back to the 0–255 intensity range.
10. Matplotlib is used to display the original and processed images together.
11. These operations demonstrate important **point processing techniques** in digital image processing.
12. Image negation, thresholding, and gamma correction can be useful for enhancement and analysis.
13. The experiment helps in understanding how pixel intensity transformations affect an image.
14. Therefore, the program successfully demonstrates basic grayscale image enhancement techniques using Python.